

Brain Tumor Detection and Localization Using YOLO v12 and Deep Learning Techniques in Python

```
# check for GPU availability
!nvidia-smi
```

Tue Feb 25 11:22:36 2025

NVIDIA-SMI 550.54.15				Driver Version: 550.54.15				CUDA Version: 12.4			
GPU	Name			Persistence-M		Bus-Id	Disp.A	Volatile	Uncorr.	ECC	
Fan	Temp	Perf		Pwr:Usage/Cap			Memory-Usage	GPU-Util	Compute	M.	
									MIG	M.	
0	Tesla T4			Off		00000000:00:04.0	Off			0	
N/A	70C	P8		11W / 70W		0MiB / 15360MiB		0%	Default	N/A	
Processes:											
GPU	GI	CI	PID	Type	Process name					GPU Memory	
	ID	ID								Usage	
No running processes found											

```
import torch
torch.cuda.empty_cache()
```

Step 01 # Install the Ultralytics Package

```
# install ultralytics
!pip install ultralytics
```

Show hidden output

Step 02 # Import All the Required Libraries

```
# import libraries
import ultralytics
ultralytics.checks()
from ultralytics import YOLO
from IPython.display import Image
```

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
Setup complete (2 CPUs, 12.7 GB RAM, 33.2/112.6 GB disk)

Step # 03 Download Dataset from Roboflow

```
# load the dataset
!pip install roboflow
```

```
from roboflow import Roboflow
rf = Roboflow(api_key="0zA9553LIS7TirJ03Cr4")
project = rf.workspace("aykut-izfoo").project("brain-tumor-detection-rwfby")
version = project.version(3)
dataset = version.download("yolov11")
```

Show hidden output

```
# check for base directory
dataset.location
```

```
!'/content/Brain-Tumor-Detection-3'
```

✓ Step # 04 Train YOLOv12 Model on a Custom Dataset

```
!yolo task=detect mode=train data={dataset.location}/data.yaml model="yolo11m.pt" epochs=20 imgsz=640 batch=8
```

```

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
12/20   7.79G    1.387    0.9343    1.569      8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.854    0.874    0.89   0.484

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
13/20   7.79G    1.367    0.9299    1.575      8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.873    0.83     0.877  0.507

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
14/20   7.79G    1.337    0.8876    1.517     10          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.881    0.843    0.871  0.516

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
15/20   7.74G    1.296    0.8434    1.486     13          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.863    0.843    0.892  0.53

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
16/20   7.79G    1.287    0.8211    1.487      8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.925    0.824    0.88   0.532

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
17/20   7.79G    1.267    0.7794    1.473      8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.911    0.874    0.921  0.552

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
18/20   7.79G    1.248    0.7439    1.46       9          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.892    0.883    0.895  0.538

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
19/20   7.74G    1.244    0.73     1.44       8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.876    0.89    0.904  0.55

Epoch  GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
20/20   7.79G    1.202    0.7193    1.413      8          640: 100% 150/150 [01:53<00:00,
      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.908    0.874    0.89   0.549

```

20 epochs completed in 0.666 hours.

Optimizer stripped from runs/detect/train/weights/last.pt, 40.5MB

Optimizer stripped from runs/detect/train/weights/best.pt, 40.5MB

Validating runs/detect/train/weights/best.pt...

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)

YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs

```

      Class  Images  Instances  Box(P      R      mAP50  mAP50-95): 100% 10/10 [00:00
      all    150      159      0.914    0.874    0.921  0.552

```

Speed: 0.3ms preprocess, 24.9ms inference, 0.1ms loss, 2.4ms postprocess per image

Results saved to **runs/detect/train**

Learn more at <https://docs.ultralytics.com/modes/train>

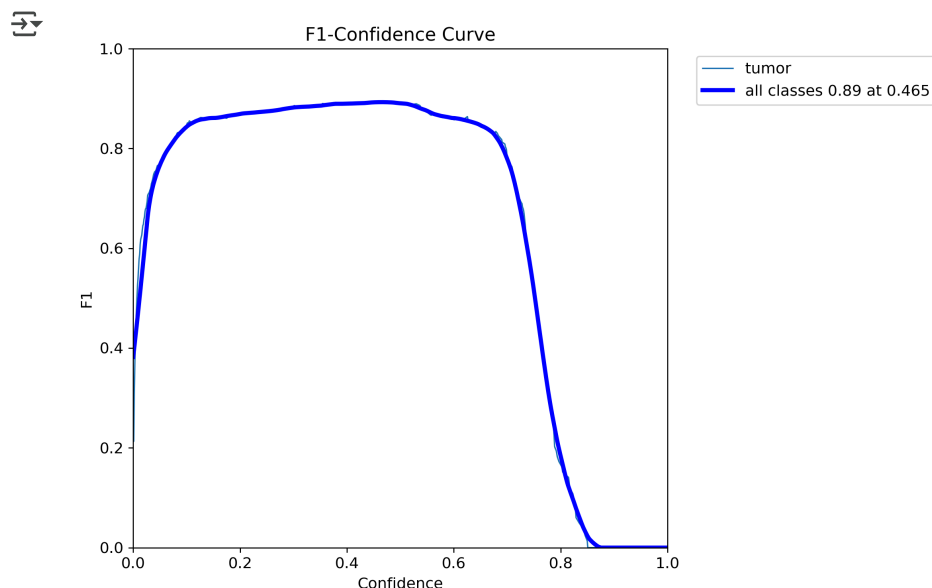
Summary

- **Training:** Completed 20 epochs in **0.668 hours**.

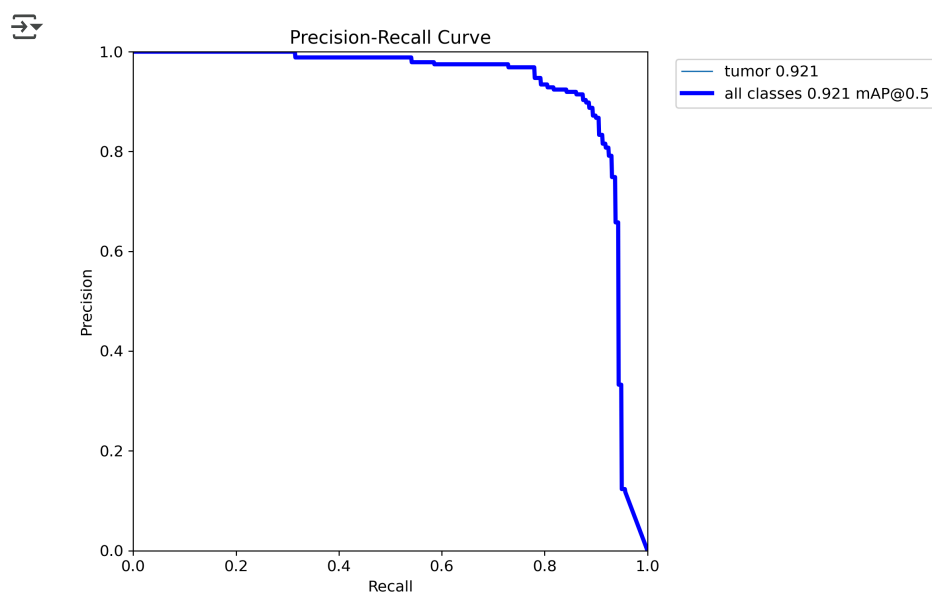
- **Model Details:** YOLOv11m with **125 layers**, **20M parameters**, and **67.6 GFLOPs**.
- **Performance Metrics:**
 - **mAP@50: 92.1%**
 - **mAP@50-95: 55.2%**
 - **Precision: 91.4%**
 - **Recall: 87.4%**
- **Speed:**
 - **Inference time: 24.9ms per image**
 - **Preprocessing: 0.4ms**
 - **Postprocessing: 3.4ms**
- **Output Saved:** runs/detect/train/weights/
- **Optimizer Removed:** Model weights (best.pt & last.pt) stripped to **40.5MB**.

✓ Step # 05 Examine Training Results

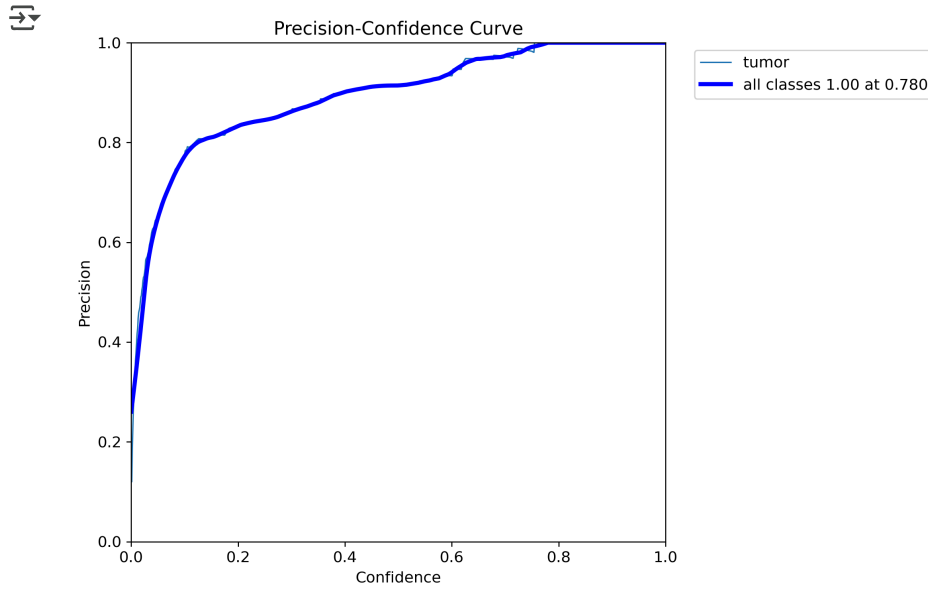
Image("/content/runs/detect/train/F1_curve.png", width=600)



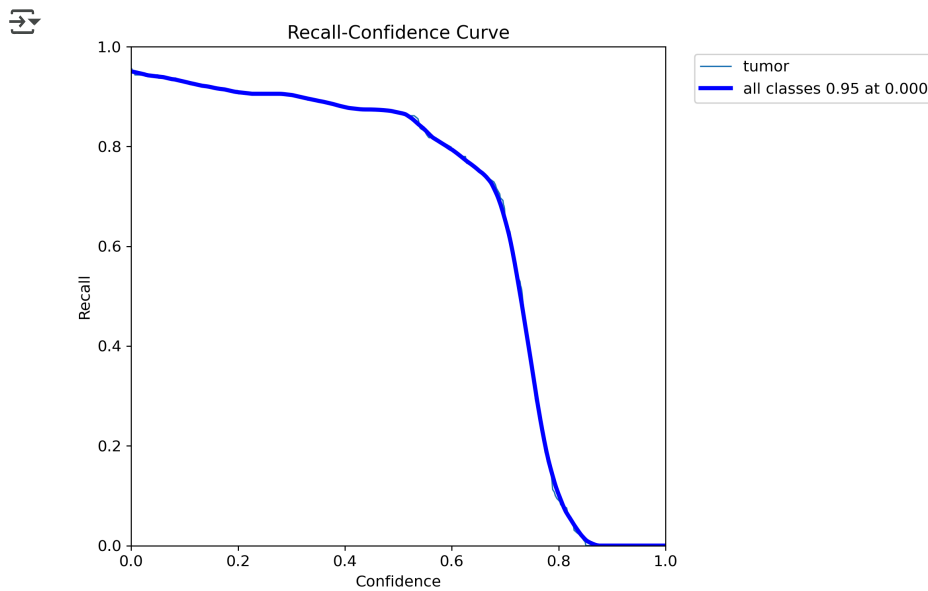
Image("/content/runs/detect/train/PR_curve.png", width=600)



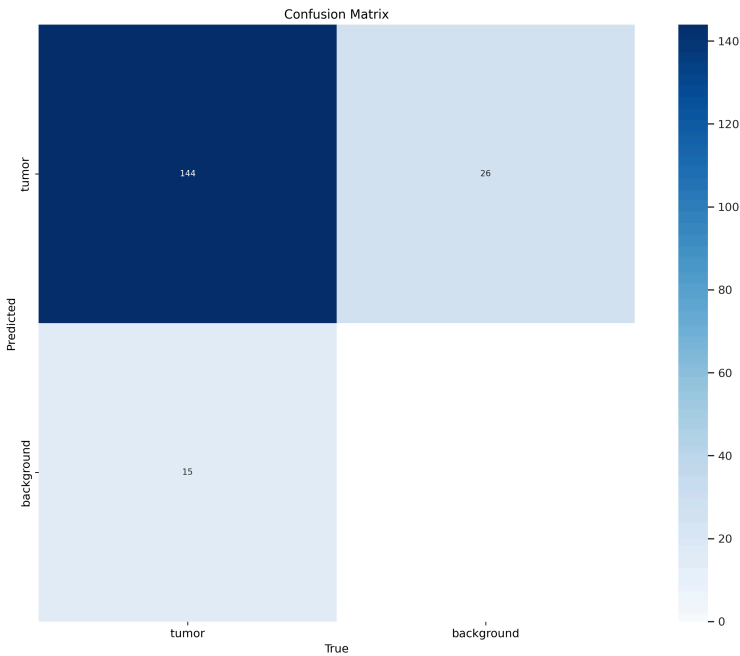
```
Image("/content/runs/detect/train/P_curve.png", width=600)
```



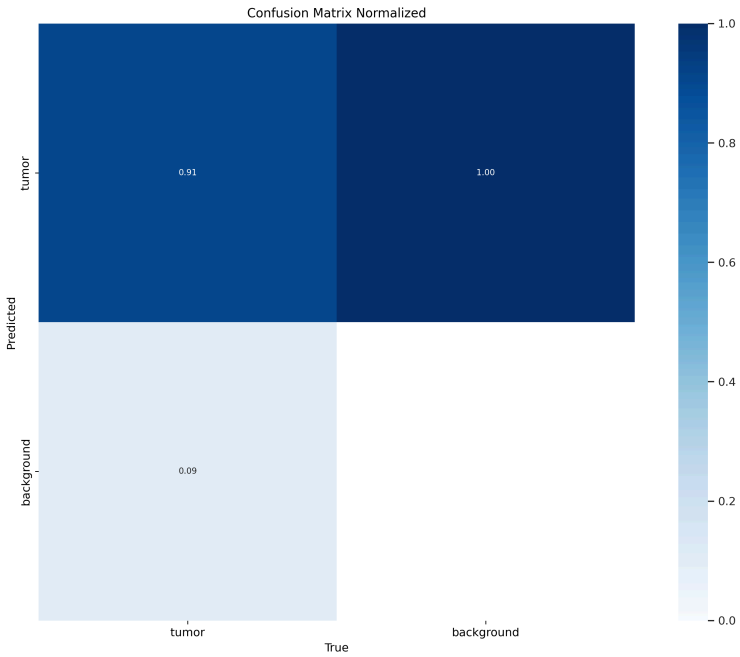
```
Image("/content/runs/detect/train/R_curve.png", width=600)
```



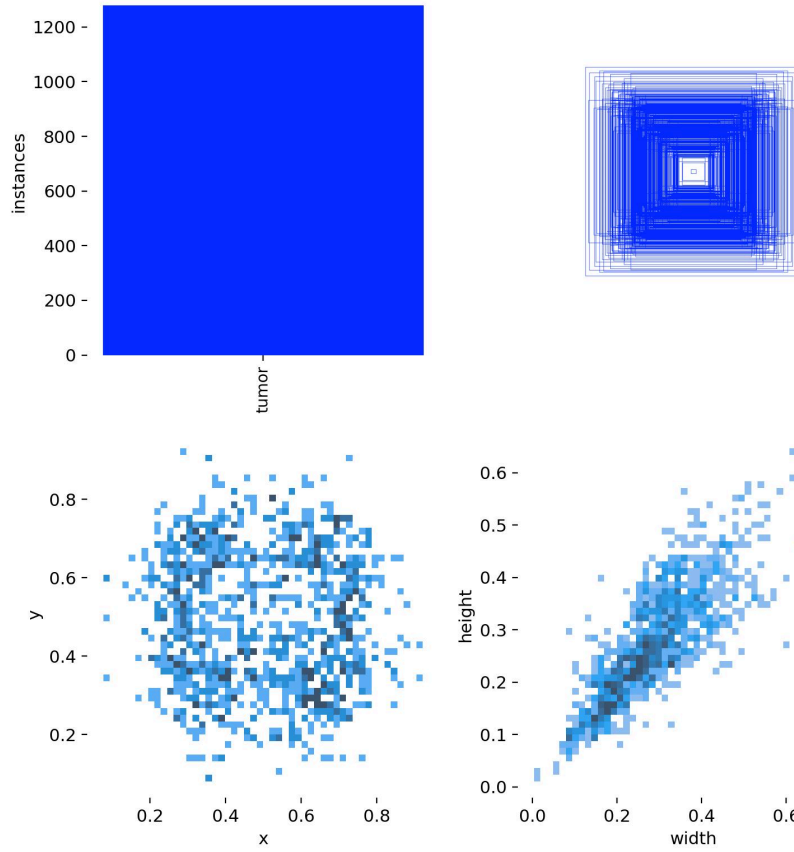
```
Image("/content/runs/detect/train/confusion_matrix.png", width=600)
```

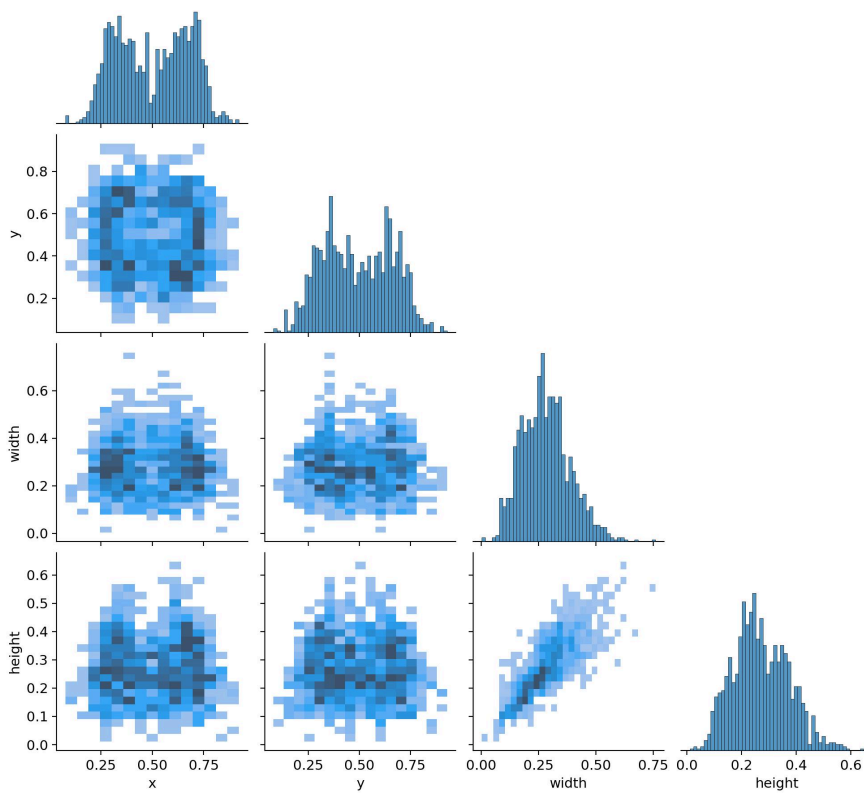
```
Image("/content/runs/detect/train/confusion_matrix_normalized.png", width=600)
```



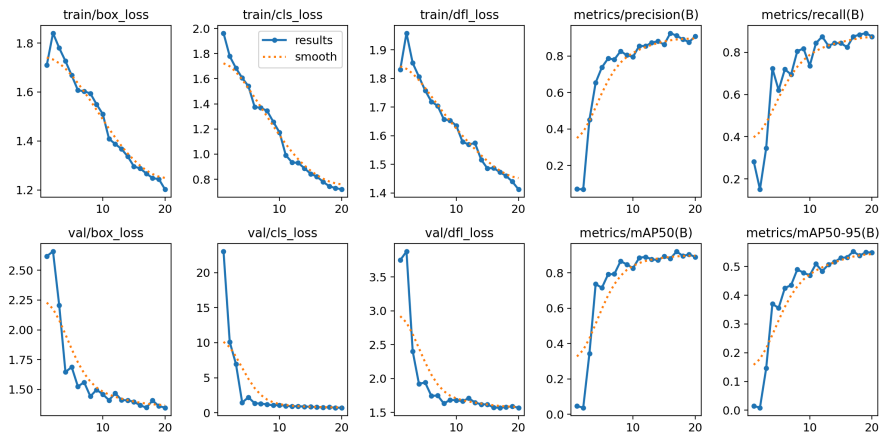
```
Image("/content/runs/detect/train/labels.jpg", width=600)
```



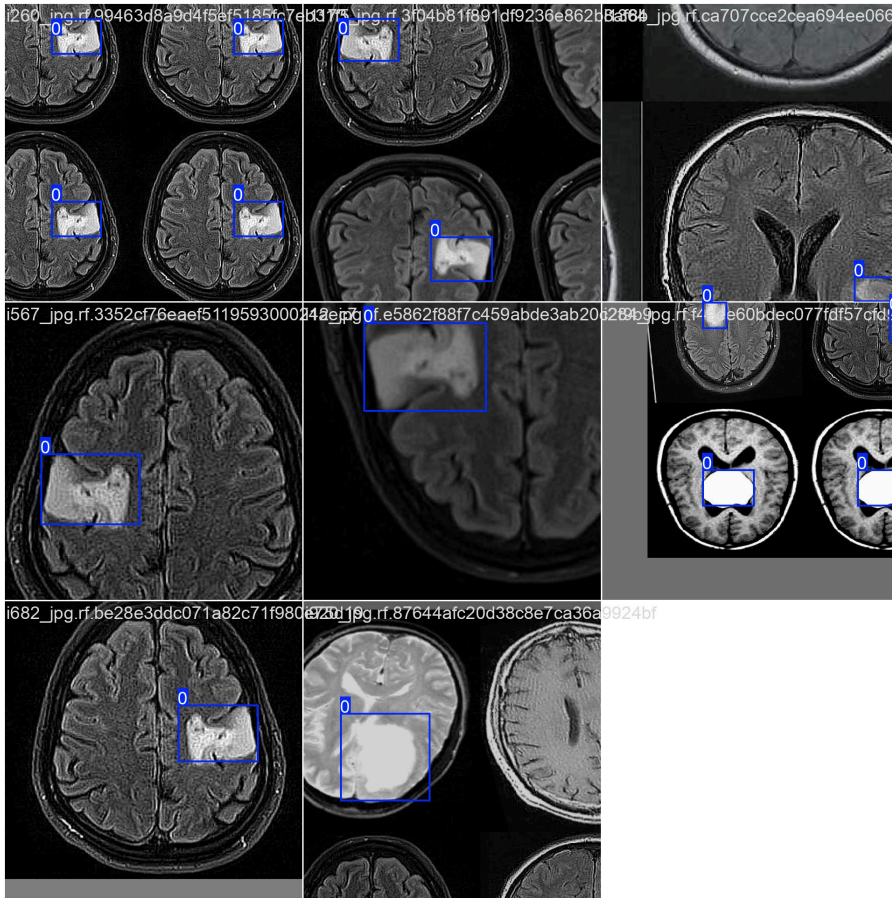
```
Image("/content/runs/detect/train/labels_correlogram.jpg", width=600)
```



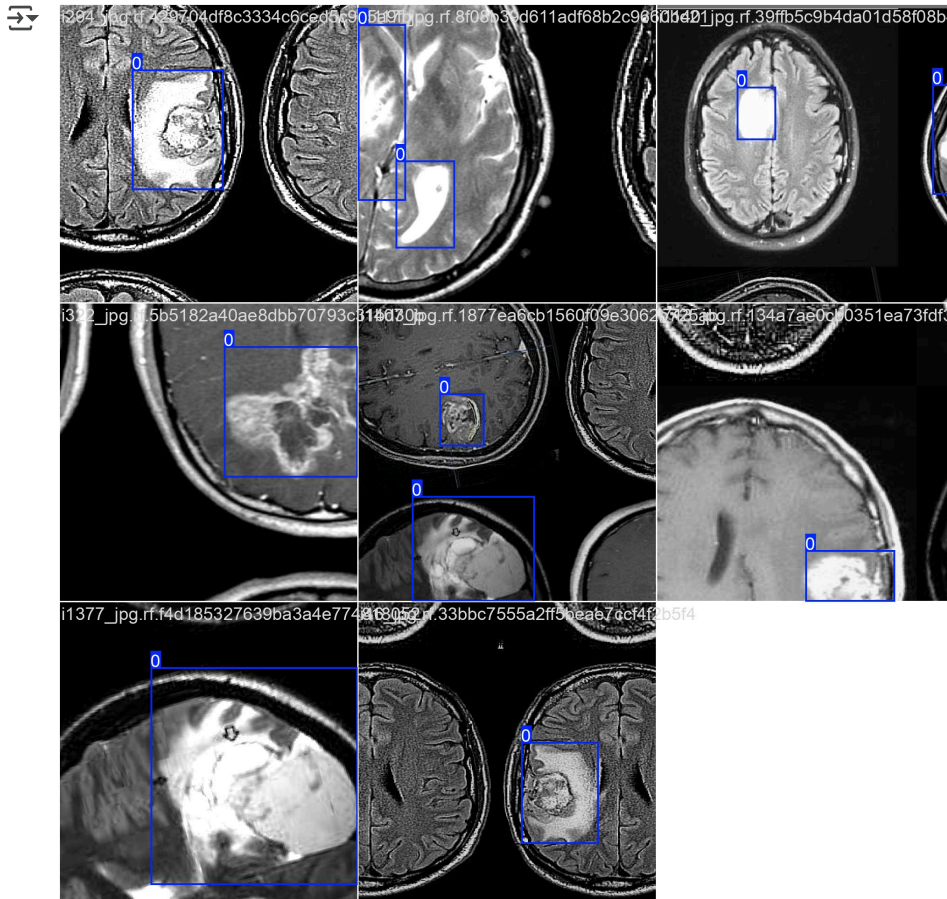
```
Image("/content/runs/detect/train/results.png", width=600)
```



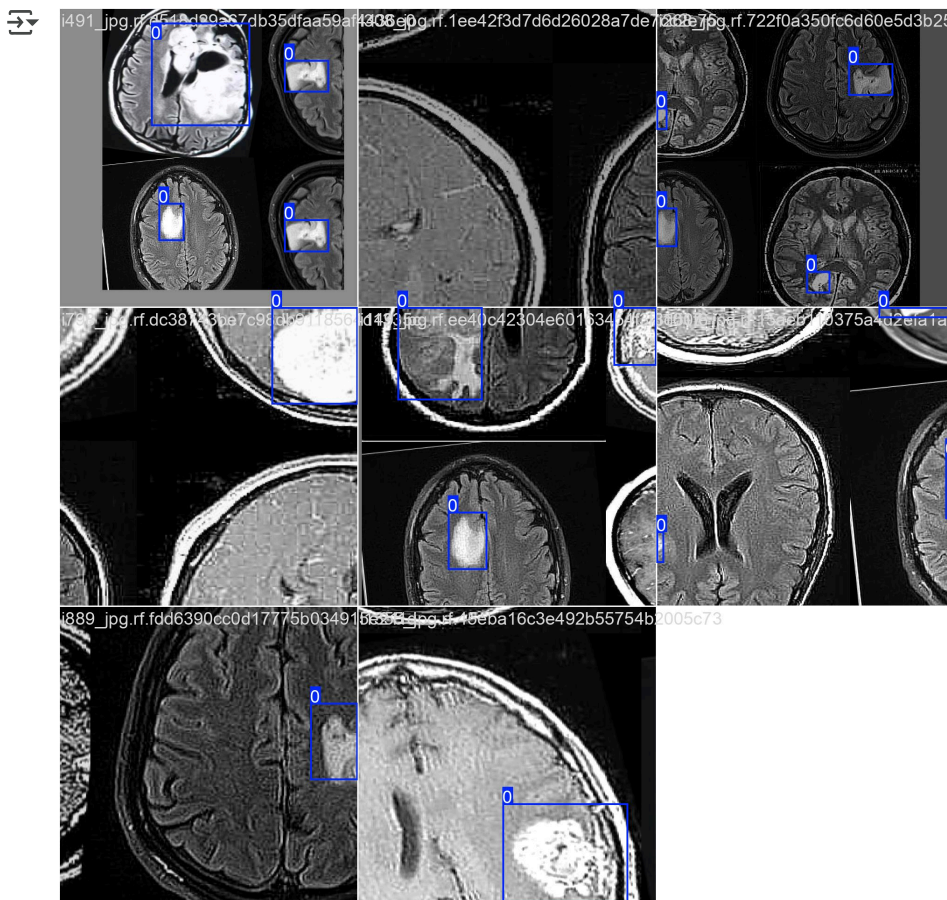
```
Image("/content/runs/detect/train/train_batch0.jpg", width=600)
```



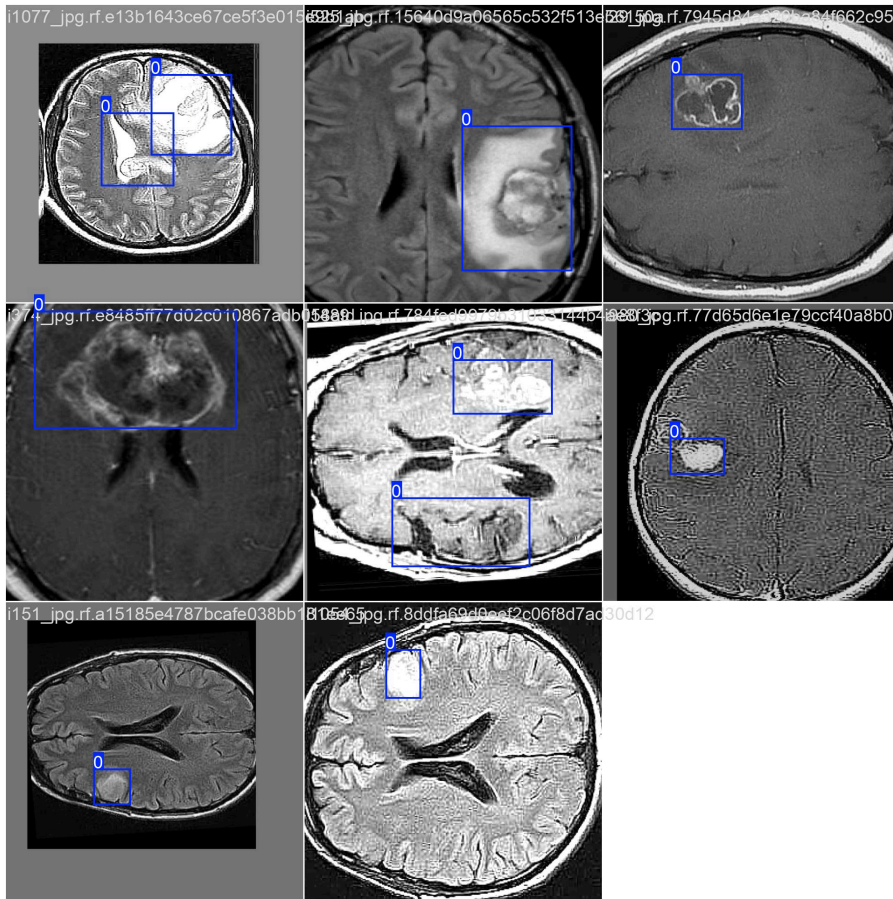
```
Image("/content/runs/detect/train/train_batch1.jpg", width=600)
```



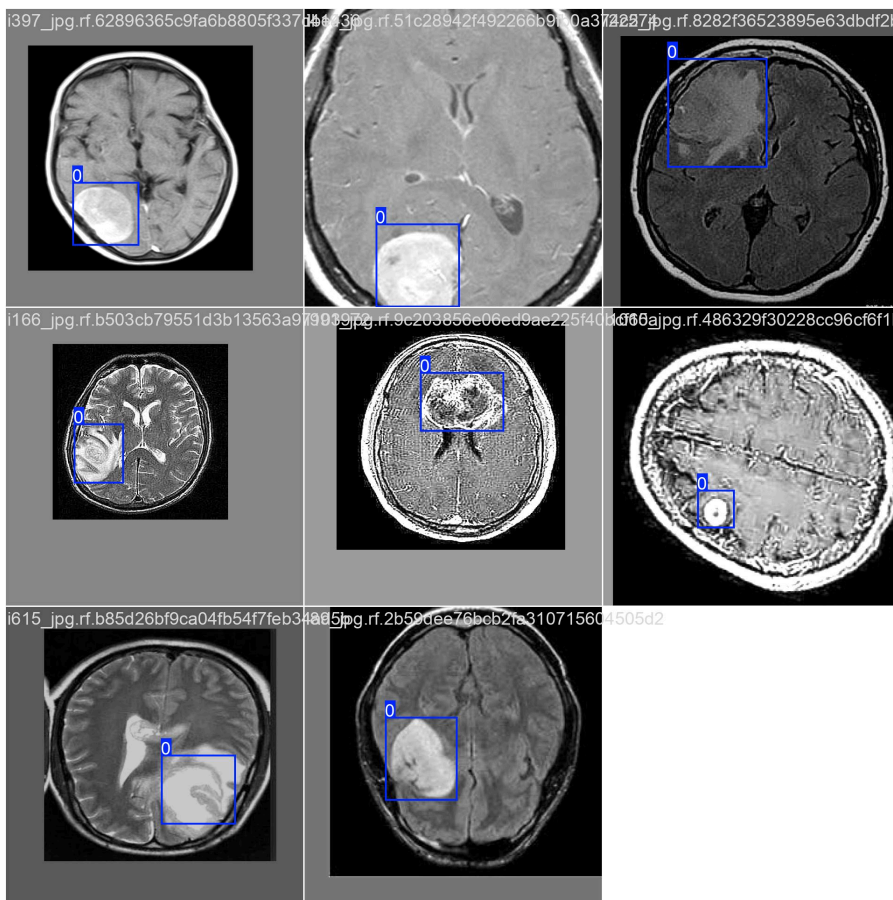
```
Image("/content/runs/detect/train/train_batch2.jpg", width=600)
```



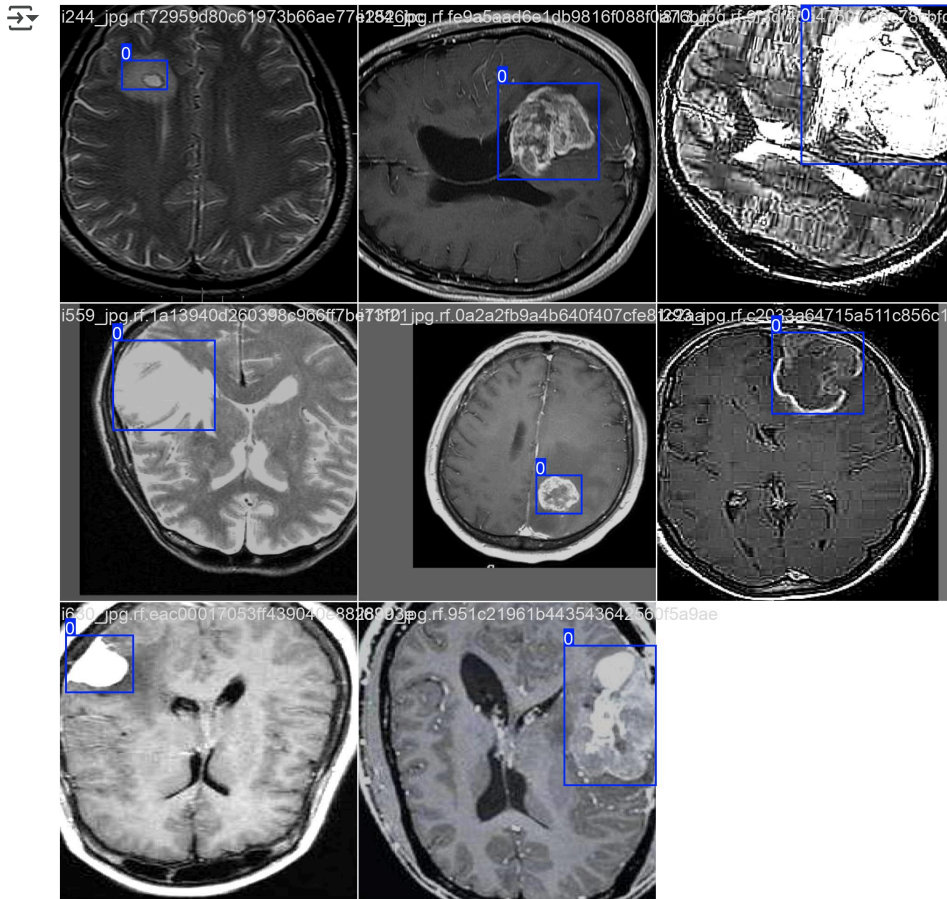
```
Image("/content/runs/detect/train/train_batch1500.jpg", width=600)
```

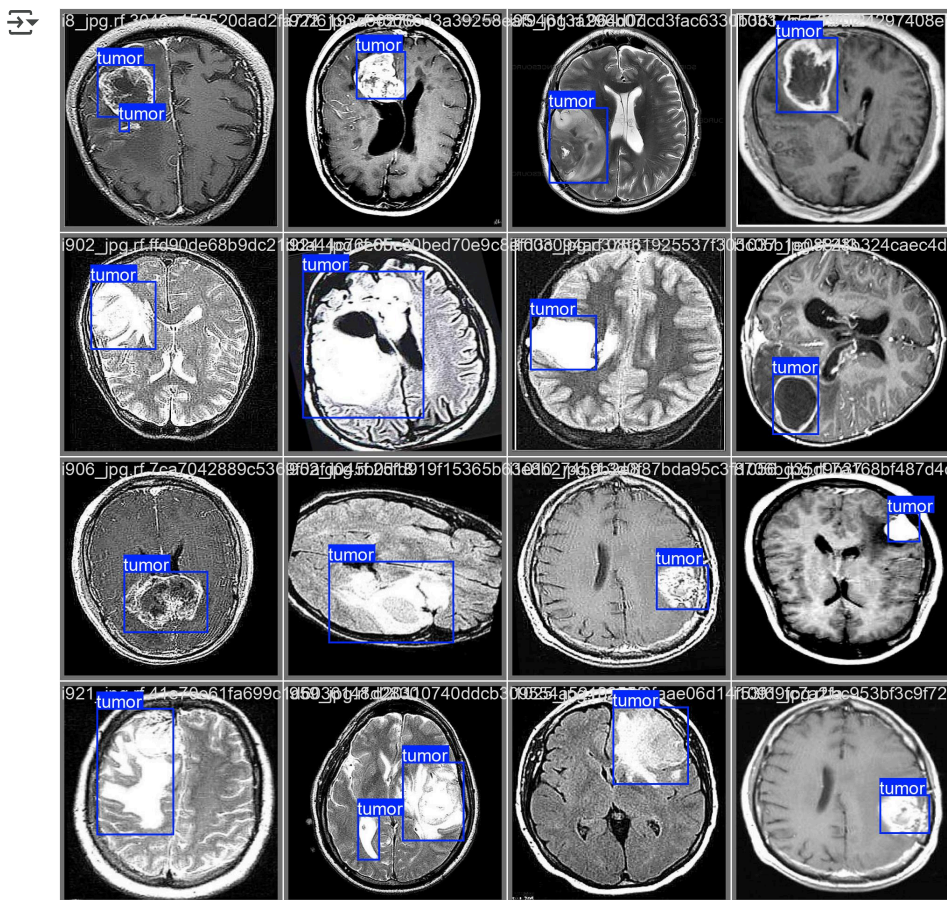
Image("/content/runs/detect/train/train_batch1501.jpg", width=600)



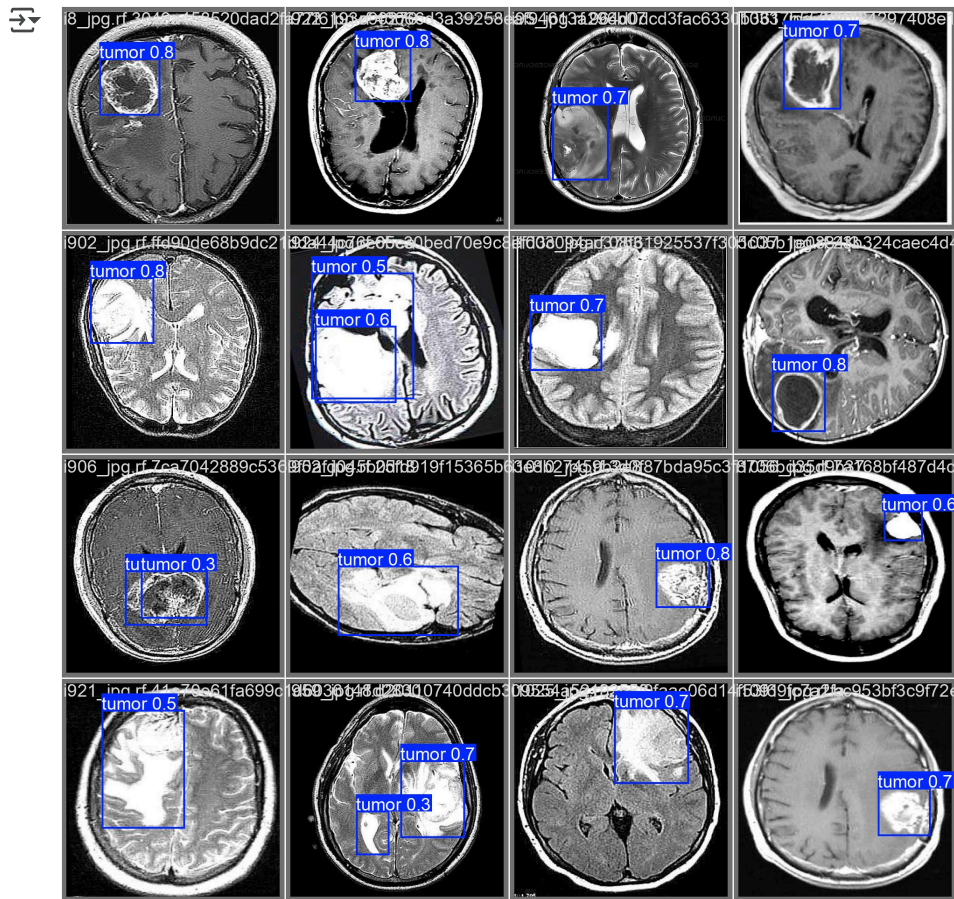
Image("/content/runs/detect/train/train_batch1502.jpg", width=600)



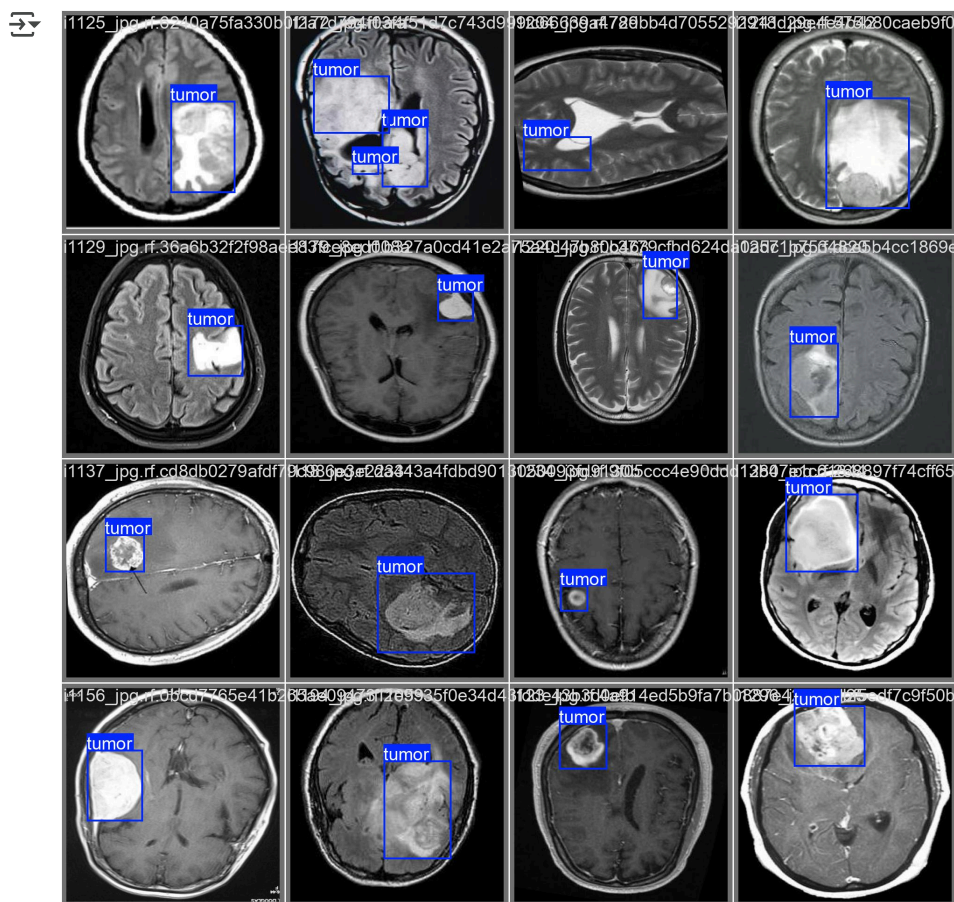
Image("/content/runs/detect/train/val_batch0_labels.jpg", width=600)



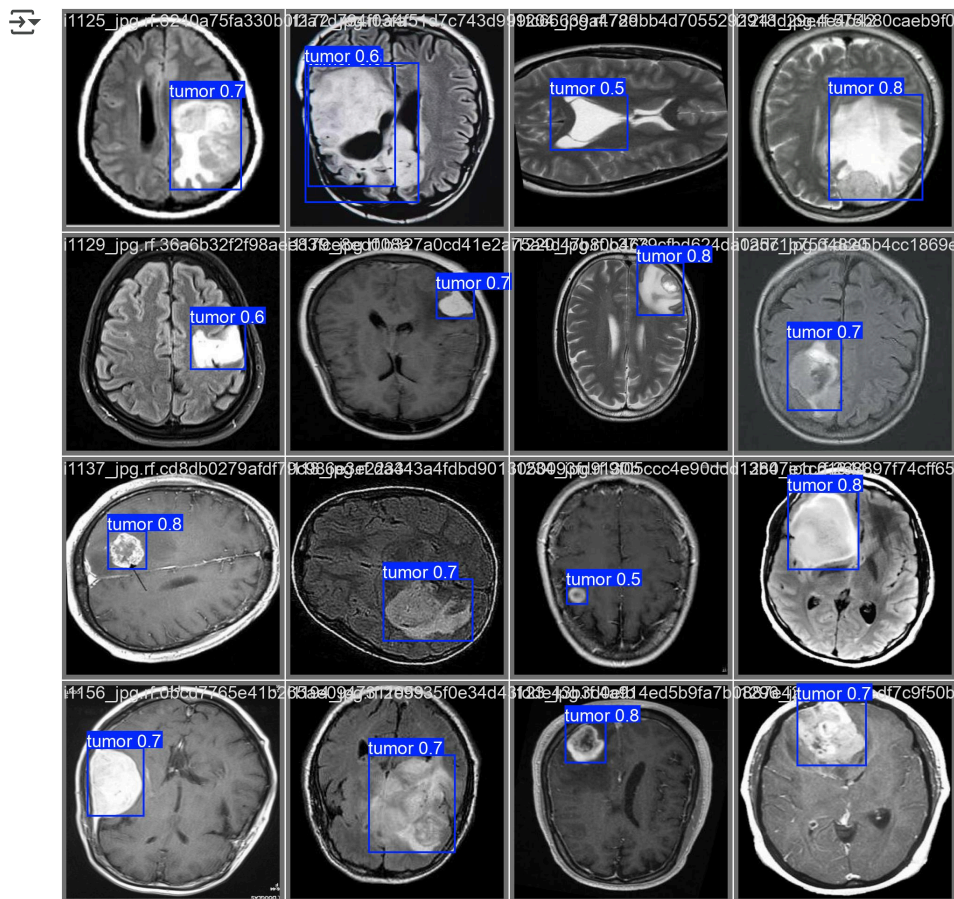
Image("/content/runs/detect/train/val_batch0_pred.jpg", width=600)



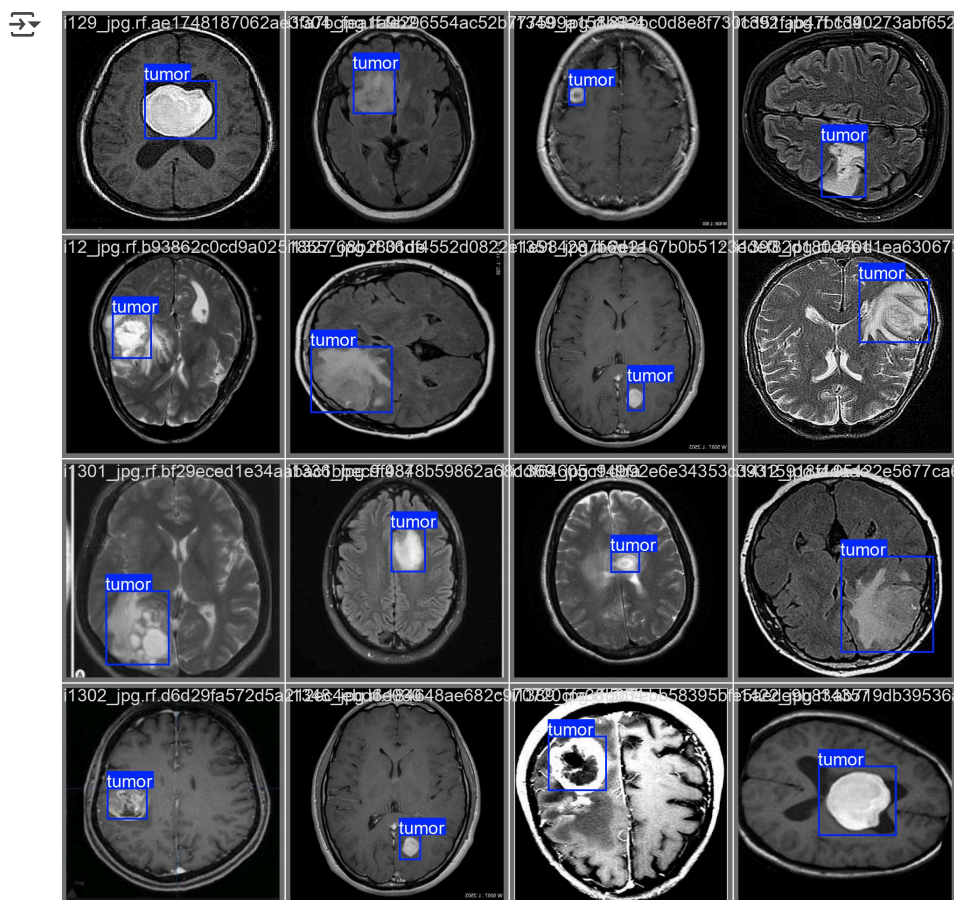
Image("/content/runs/detect/train/val_batch1_labels.jpg", width=600)



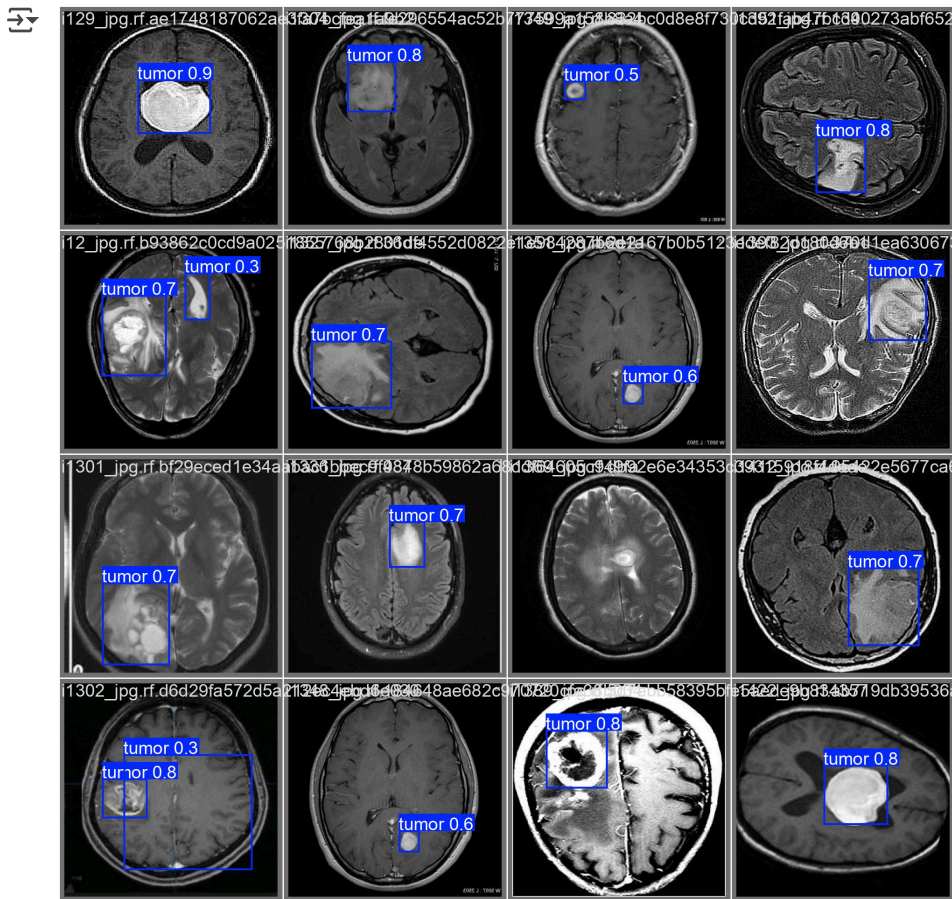
Image("/content/runs/detect/train/val_batch1_pred.jpg", width=600)



Image("/content/runs/detect/train/val_batch2_labels.jpg", width=600)



Image("/content/runs/detect/train/val_batch2_pred.jpg", width=600)



✓ Step # 06 Validate Fine-Tuned Model

```
!yolo task=detect mode=val model="/content/runs/detect/train/weights/last.pt" data={dataset.location}/data.yaml
```

Ultralytics 8.3.78 Python-3.11.11 torch-2.5.1+cu124 CUDA:0 (Tesla T4, 15095MiB)
 YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs
val: Scanning /content/Brain-Tumor-Detection-3/valid/labels.cache... 150 images, 0 backgrounds, 0 corrupt:

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95)
all	150	159	0.908	0.874	0.89	0.549

 Speed: 4.2ms preprocess, 23.7ms inference, 0.0ms loss, 2.0ms postprocess per image
 Results saved to **runs/detect/val**
 Learn more at <https://docs.ultralytics.com/modes/val>

Validation Summary

- **Model:** YOLOv11m (Ultralytics 8.3.78)
- **Hardware:** Tesla T4 (CUDA)
- **Dataset:** 150 images, 159 instances, no corrupt data
- **Performance Metrics:**
 - **mAP@50:** 89.0%
 - **mAP@50-95:** 54.9%
 - **Precision:** 90.8%
 - **Recall:** 87.4%
- **Speed:**
 - **Inference:** 25.5ms per image
 - **Preprocessing:** 0.3ms
 - **Postprocessing:** 2.5ms
- **Results Saved:** runs/detect/val/

✓ Step # 07 Inference with Custom Model on Images

```
!yolo task=detect mode=predict model="/content/runs/detect/train/weights/best.pt" conf=0.25 source={dataset.lo
```

```

image 96/150 /content/Brain-Tumor-Detection-3/test/images/i588_jpg.rf.6b3391fa86a9d4aff82a8f521391368c.jpg
image 97/150 /content/Brain-Tumor-Detection-3/test/images/i58_jpg.rf.8506a4568133f10cd507adef43982620.jpg:
image 98/150 /content/Brain-Tumor-Detection-3/test/images/i623_jpg.rf.9bdcabcabfc5e51bd5a2724771e80d98.jpg
image 99/150 /content/Brain-Tumor-Detection-3/test/images/i632_jpg.rf.297489cc9d56e7d262e83545e977ea7a.jpg
image 100/150 /content/Brain-Tumor-Detection-3/test/images/i637_jpg.rf.46c2211bedfd6f6cedfd715855c640da.jpg
image 101/150 /content/Brain-Tumor-Detection-3/test/images/i638_jpg.rf.76b6088d4ba38314b4372a6d2efbca3b.jpg
image 102/150 /content/Brain-Tumor-Detection-3/test/images/i641_jpg.rf.a4954f7f47ec8613bae4a57cbce730e0.jpg
image 103/150 /content/Brain-Tumor-Detection-3/test/images/i653_jpg.rf.e4e6062de8130ad6d2038151f592f23a.jpg
image 104/150 /content/Brain-Tumor-Detection-3/test/images/i666_jpg.rf.23aa2b1551189b8ad68e77b03d12f545.jpg
image 105/150 /content/Brain-Tumor-Detection-3/test/images/i669_jpg.rf.4e56b286ed4308928dbd96b50c25ef3e.jpg
image 106/150 /content/Brain-Tumor-Detection-3/test/images/i675_jpg.rf.224e3e23da129a4a0b5b3a098dd16ec4.jpg
image 107/150 /content/Brain-Tumor-Detection-3/test/images/i683_jpg.rf.b2dfef6ca9297f9364b618e982144f75.jpg
image 108/150 /content/Brain-Tumor-Detection-3/test/images/i694_jpg.rf.13e27026ee0030b9053d430d68152c1a.jpg
image 109/150 /content/Brain-Tumor-Detection-3/test/images/i712_jpg.rf.567d7079fdbc99d48d17867278527d8.jpg
image 110/150 /content/Brain-Tumor-Detection-3/test/images/i724_jpg.rf.4c51b2c0c5719bdb454f02d9829a58f8.jpg
image 111/150 /content/Brain-Tumor-Detection-3/test/images/i726_jpg.rf.a5664c07c58928c8d2c9574ca752a33d.jpg
image 112/150 /content/Brain-Tumor-Detection-3/test/images/i734_jpg.rf.099f749013745047d94661a66722a9a9.jpg
image 113/150 /content/Brain-Tumor-Detection-3/test/images/i737_jpg.rf.aff5ce741a41fc19e86202e7e31f82c3.jpg
image 114/150 /content/Brain-Tumor-Detection-3/test/images/i747_jpg.rf.5f606b9859adebf1af26ed621f4bb592.jpg
image 115/150 /content/Brain-Tumor-Detection-3/test/images/i748_jpg.rf.b17f73f2c20648630ace8d65018b0aa1.jpg
image 116/150 /content/Brain-Tumor-Detection-3/test/images/i758_jpg.rf.e88ebfc5c4dc3ca39f347a8fee35ac87.jpg
image 117/150 /content/Brain-Tumor-Detection-3/test/images/i761_jpg.rf.02f776ba1f0d601e560792735cdc40b0.jpg
image 118/150 /content/Brain-Tumor-Detection-3/test/images/i771_jpg.rf.4207b274e51e172c643ea7948cfea397.jpg
image 119/150 /content/Brain-Tumor-Detection-3/test/images/i789_jpg.rf.ea9b8127a52fb24ea03907e4fbfc4147.jpg
image 120/150 /content/Brain-Tumor-Detection-3/test/images/i805_jpg.rf.a955bbb1b73ab245524d89ced3a0563f.jpg
image 121/150 /content/Brain-Tumor-Detection-3/test/images/i807_jpg.rf.314992ca981c655d2eea5b7cfe2254de.jpg
image 122/150 /content/Brain-Tumor-Detection-3/test/images/i819_jpg.rf.cc3e33a1726054de9372342b2ae4eec8.jpg
image 123/150 /content/Brain-Tumor-Detection-3/test/images/i827_jpg.rf.324fc10d039b9bb131b342def5b873a.jpg
image 124/150 /content/Brain-Tumor-Detection-3/test/images/i828_jpg.rf.62524a984eaa937c24d2f2c5c66ed3a19.jpg
image 125/150 /content/Brain-Tumor-Detection-3/test/images/i82_jpg.rf.b3eb7b97647849aca8c8cfaac3bdd991.jpg
image 126/150 /content/Brain-Tumor-Detection-3/test/images/i833_jpg.rf.4e9845b2514a319c89faca07cf68b279.jpg
image 127/150 /content/Brain-Tumor-Detection-3/test/images/i843_jpg.rf.a59ba973eeb237c05055358e26c12472.jpg
image 128/150 /content/Brain-Tumor-Detection-3/test/images/i846_jpg.rf.d24a8bf1cc973173c469ea8e72a4c14f.jpg
image 129/150 /content/Brain-Tumor-Detection-3/test/images/i857_jpg.rf.02ee23f4c9397045a1c3d960b79b24b5.jpg
image 130/150 /content/Brain-Tumor-Detection-3/test/images/i868_jpg.rf.c8a0b2fe11eb41a75b18ad55d212fd89.jpg
image 131/150 /content/Brain-Tumor-Detection-3/test/images/i884_jpg.rf.4fcd40fe90b3926fea7294c4ffb1e61c.jpg
image 132/150 /content/Brain-Tumor-Detection-3/test/images/i896_jpg.rf.4c5b309df255e568836699b08e8e34e.jpg
image 133/150 /content/Brain-Tumor-Detection-3/test/images/i908_jpg.rf.12514bd53bedd5b736af703e79f3f1be.jpg
image 134/150 /content/Brain-Tumor-Detection-3/test/images/i910_jpg.rf.33913149831b29dd515fdbcb9446edb1c.jpg
image 135/150 /content/Brain-Tumor-Detection-3/test/images/i912_jpg.rf.8f72c97302d1dbc224597f796c61a8ca.jpg
image 136/150 /content/Brain-Tumor-Detection-3/test/images/i926_jpg.rf.861915433eedb4352757f3d0d6b2b6db.jpg
image 137/150 /content/Brain-Tumor-Detection-3/test/images/i935_jpg.rf.95016f5f65086d3a8c3e2a62672bed1a.jpg
image 138/150 /content/Brain-Tumor-Detection-3/test/images/i941_jpg.rf.08a84ea2cc0abaad980adb41ff4bf342.jpg
image 139/150 /content/Brain-Tumor-Detection-3/test/images/i965_jpg.rf.52bd1c05abbcb6fe082cfb8c5a70b3f36.jpg
image 140/150 /content/Brain-Tumor-Detection-3/test/images/i969_jpg.rf.c441e283f4968ef217dfd515d88b3e3e.jpg
image 141/150 /content/Brain-Tumor-Detection-3/test/images/i96_jpg.rf.7ca0d84fc0783bba267f125e4ed1c620.jpg
image 142/150 /content/Brain-Tumor-Detection-3/test/images/i970_jpg.rf.174ea2ccc37f0523214af069eb705615.jpg
image 143/150 /content/Brain-Tumor-Detection-3/test/images/i983_jpg.rf.eb2a296d825cde6962d76a841d0f5d12.jpg
image 144/150 /content/Brain-Tumor-Detection-3/test/images/i987_jpg.rf.3f5bf6959857deed8375852d6d977f2d.jpg
image 145/150 /content/Brain-Tumor-Detection-3/test/images/i988_jpg.rf.a30953ab658ccbd589a9c95987ca4650.jpg
image 146/150 /content/Brain-Tumor-Detection-3/test/images/i98_jpg.rf.b0c828ac25c2196641290302201bcf6f.jpg
image 147/150 /content/Brain-Tumor-Detection-3/test/images/i990_jpg.rf.4bce0a3e80e3ef872db7d8c055718634.jpg
image 148/150 /content/Brain-Tumor-Detection-3/test/images/i992_jpg.rf.cffa34774438e2aee449765a6a680757.jpg
image 149/150 /content/Brain-Tumor-Detection-3/test/images/i993_jpg.rf.e5ba6e0bcae9fc7be2ee42f17981eb18.jpg
image 150/150 /content/Brain-Tumor-Detection-3/test/images/i995_jpg.rf.2fdc64de1cf961e2344e4f9f42961e29.jpg
Speed: 2.1ms preprocess, 20.3ms inference, 2.2ms postprocess per image at shape (1, 3, 640, 640)
Results saved to runs/detect/predict

```

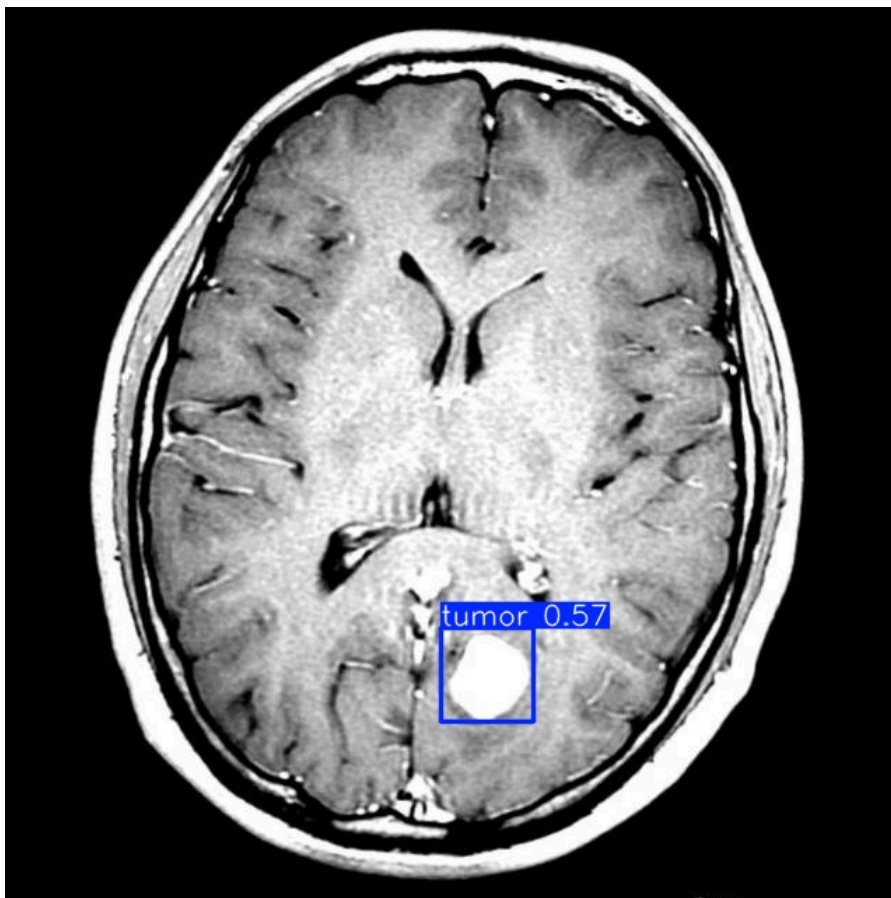
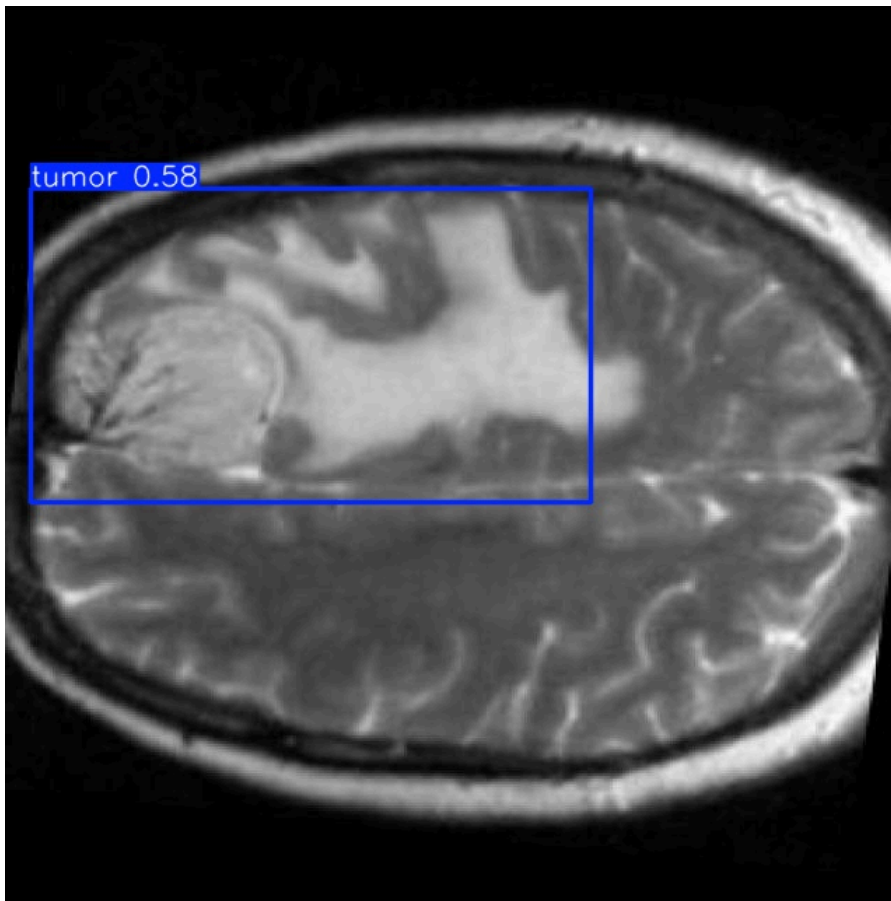
🔗 Learn more at <https://docs.ultralytics.com/modes/predict>

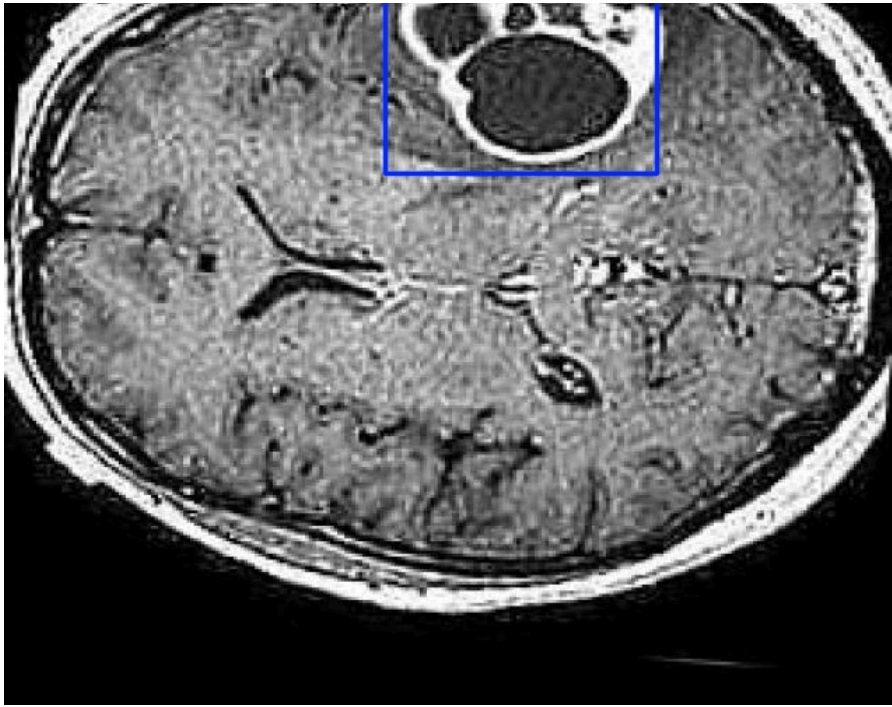
```

import glob
import os
from IPython.display import Image as IPyImage, display

latest_folder = max(glob.glob('/content/runs/detect/predict*/*'), key=os.path.getmtime)
for img in glob.glob(f'{latest_folder}/*.jpg')[1:4]:
    display(IPyImage(filename=img, width=600))
    print("\n")

```





```
# convert model to torchscript
model = YOLO("/content/runs/detect/train/weights/best.pt")
model.export(format="torchscript")
```

🔗 Ultralytics 8.3.78 🚀 Python-3.11.11 torch-2.5.1+cu124 CPU (Intel Xeon 2.00GHz)
YOLO11m summary (fused): 125 layers, 20,030,803 parameters, 0 gradients, 67.6 GFLOPs

PyTorch: starting from '/content/runs/detect/train/weights/best.pt' with input shape (1, 3, 640, 640) BCHW

TorchScript: starting export with torch 2.5.1+cu124...

TorchScript: export success ✅ 7.4s, saved as '/content/runs/detect/train/weights/best.torchscript' (77.1 MB)

Export complete (10.9s)

Results saved to /content/runs/detect/train/weights

Predict: yolo predict task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=640

Validate: yolo val task=detect model=/content/runs/detect/train/weights/best.torchscript imgsz=640

Visualize: <https://netron.app>

'/content/runs/detect/train/weights/best.torchscript'

```
import cv2
import torch
import matplotlib.pyplot as plt
from ultralytics import YOLO
```

```
def predict_and_display_v11(image_path, model_path, confidence_threshold=0.5, target_size=(640, 640)):
    """
```

Detects and visualizes brain tumor presence in an image using a fine-tuned YOLOv11 model.
Enhances the image by resizing and converting it to grayscale before prediction.

Args:

image_path: Path to the input image.

model_path: Path to the trained YOLOv11 TorchScript model.

confidence_threshold: Minimum confidence score for a detection to be considered.

target_size: The target size for resizing the image.

"""

Load and preprocess the image

img = cv2.imread(image_path)

img_resized = cv2.resize(img, target_size)